THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE TWO-CHAPTER TEST SET C

Chapters 5 & 6: **Describing Motion Around Us, How Forces Affect Motion** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A cyclist covers 1500 m in 100 s along a straight road without turning back. The average speed is: 1

(a) 7.5 m s^{-1} **(b)** 30 m s^{-1} **(c)** 15 m s^{-1} **(d)** 0.07 m s^{-1}

2. The velocity of a car changes from 0 m s^{-1} to 20 m s^{-1} in 5 s in a straight line. Its average acceleration is: 1

(a) −4 m s^{-2} **(b)** 8 m s^{-2} **(c)** 5 m s^{-2} **(d)** 4 m s^{-2}

3. A body with initial velocity 0 m s^{-1} moves with constant acceleration 3 m s^{-2} for 6 s. Its final velocity is: 1

(a) 9 m s^{-1} **(b)** 3 m s^{-1} **(c)** 18 m s^{-1} **(d)** 36 m s^{-1}

4. A body starts with velocity 10 m s^{-1} and accelerates uniformly at 2 m s^{-2} for 4 s. The distance travelled is: 1

(a) 112 m **(b)** 48 m **(c)** 32 m **(d)** 56 m

5. A vehicle moving at 10 m s^{-1} is brought to rest with a uniform retardation of 2 m s^{-2}. The stopping distance is: 1

(a) 50 m **(b)** 12.5 m **(c)** 25 m **(d)** 5 m

6. The net force needed to give a body of mass 50 kg an acceleration of 2 m s^{-2} is: 1

(a) 52 N **(b)** 100 N **(c)** 25 N **(d)** 200 N

7. A net force of 12 N acts on a body of mass 8 kg. Its acceleration is: 1

(a) 3 m s^{-2} **(b)** 96 m s^{-2} **(c)** 0.67 m s^{-2} **(d)** 1.5 m s^{-2}

8. Two forces of 10 N and 6 N act on a block along the same straight line in opposite directions. The magnitude of the net force is: 1

(a) 10 N **(b)** 60 N **(c)** 4 N **(d)** 16 N

9. A force of 60 N produces an acceleration of 3 m s^{-2} in a body. Its mass is: 1

(a) 20 kg **(b)** 180 kg **(c)** 3 kg **(d)** 40 kg

10. The SI unit of force is: 1

(a) pascal **(b)** joule **(c)** watt **(d)** newton

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Why is circular motion at constant speed accelerated? 2

12. What is the direction of velocity in uniform circular motion? 2

13. Calculate the weight of a 30 kg body ($g = 10\ m\ s^{-2}$). 2

14. What are balanced forces? Give an example. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 30–40 words each.

15. A stone is dropped from a tower and takes 3 s to hit the ground. Find the height and final velocity ($g = 10\ m\ s^{-2}$). 3

16. Find the force on a 1500 kg sports car accelerating from rest to 10 m s^{-1} in 5 s. What force acts when it later comes to rest in 5 s? 3

17. Two cars A and B have position–time lines of slopes 10 and 20 m s^{-1}. Which is faster? Find the distance each covers in 5 s. 3

OR

Is a scooter with a constant speedometer reading always unaccelerated? Explain. 3

18. Out of 100 oarsmen in a boat, 95 row backwards and 5 row forwards, each applying 200 N. Find the net force on the boat. 3

OR

Why is it said that forces always occur in pairs? Give two examples. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. (i) Distinguish scalar and vector with two examples each from this chapter.
(ii) List four differences between distance and displacement. 5

OR

A train of speed 90 km h^{-1} is brought to rest in 50 s. 5
(i) Convert the speed.
(ii) Find retardation.
(iii) Find the stopping distance.
(iv) What fraction of the distance is covered in the first 25 s?

20. **Case study:** A weightlifter holds a barbell of total mass 30 kg ($g = 9.8\ m\ s^{-2}$). 5
(i) What forces act on it?
(ii) Find the weight.
(iii) Find the force she applies.
(iv) What would be the net force?

OR

(i) Differentiate mass and weight. 5
(ii) What is the weight of a 60 kg person on the Earth (g = 10)?
(iii) On the Moon where g = 1.6?

END OF QUESTION PAPER — **Total: 40 Marks**